

# AI-POWERED CITIZEN CALL INTELLIGENCE PLATFORM

## Department Master Data, Complaint Taxonomy & Metadata Specification

---

### 1. PURPOSE OF THIS DOCUMENT

This document defines the **department structure, complaint categories, subcategories, metadata fields, SLA rules, priority rules, routing logic, officer metadata, location metadata, AI metadata, and analytics metadata** required for the proposed **AI-Powered Citizen Call Intelligence Platform**.

The objective is to create a standardized data structure so that every incoming citizen call can be converted from an unstructured conversation into a **structured, trackable and actionable complaint record**.

The core transformation is:

**Citizen Voice → Transcript → AI Understanding → Structured Metadata → Department Routing → SLA → Resolution → Analytics**

---

### 2. PROPOSED GOVERNMENT DEPARTMENT MASTER

For the hackathon prototype, the platform can support the following **12 major departments**.

| ID   | Department                    | Main Responsibility                                  |
|------|-------------------------------|------------------------------------------------------|
| D001 | Water Supply & Sewerage       | Water supply, leakage, drainage, sewage              |
| D002 | Sanitation & Solid Waste      | Garbage, waste collection, cleanliness               |
| D003 | Roads & Public Works          | Roads, potholes, bridges, public infrastructure      |
| D004 | Electricity & Street Lighting | Power issues and streetlights                        |
| D005 | Public Transport              | Buses, bus stops, transport complaints               |
| D006 | Traffic Management            | Traffic congestion, signals, road safety             |
| D007 | Public Health                 | Public health, sanitation hazards, health facilities |

| ID   | Department                      | Main Responsibility                                           |
|------|---------------------------------|---------------------------------------------------------------|
| D008 | Emergency & Disaster Management | Floods, fires, disasters, major emergencies                   |
| D009 | Police & Public Safety          | Crime, safety, law-and-order emergencies                      |
| D010 | Municipal Services              | Civic services, public spaces, encroachments                  |
| D011 | Environment                     | Pollution, environmental complaints, trees                    |
| D012 | Revenue & Civic Administration  | Property tax, certificates, municipal administrative services |

For a 12-hour hackathon prototype, **12 departments are enough**. Do not create 30–40 departments because it will make your routing and metadata unnecessarily complicated.

## 3. DEPARTMENT D001 — WATER SUPPLY & SEWERAGE

### Department Metadata

**Department ID:** D001

**Department Name:** Water Supply & Sewerage

**Department Code:** WATER

**Parent Domain:** Public Utilities

**Primary Function:** Ensure reliable water supply, drainage and sewage management.

### Complaint Categories

#### A. Water Supply

Subcategories:

- No water supply
- Low water pressure
- Irregular water supply
- Contaminated water
- Water supply timing issue
- Water tanker request
- Pipeline disruption

- New water connection

## B. Water Leakage

- Pipeline leakage
- Roadside water leakage
- Main pipeline burst
- Household connection leakage

## C. Sewerage

- Sewage overflow
- Blocked drainage
- Open sewer
- Sewer leakage
- Bad sewage smell

## Common Keywords

### English:

water, pipeline, leakage, tanker, drainage, sewage, dirty water, no supply, pressure

### Tamil/Tanglish examples:

thanni, water varala, thanni varala, pipe leak, drainage, saakkadai, sewage, tanker

## Suggested SLA

| Priority | Example                                   | SLA      |
|----------|-------------------------------------------|----------|
| Critical | Major pipeline burst affecting large area | 4 hours  |
| High     | No water supply                           | 12 hours |
| Medium   | Low pressure                              | 24 hours |
| Low      | New connection inquiry                    | 72 hours |

# 4. DEPARTMENT D002 — SANITATION & SOLID WASTE

**Department ID:** D002

**Department Code:** SANITATION

## Complaint Categories

### Waste Collection

- Garbage not collected
- Irregular collection
- Overflowing garbage bins
- Missed collection

### Public Cleanliness

- Dirty streets
- Public area cleanliness
- Illegal dumping
- Dead animal disposal

### Waste Management

- Construction waste
- Plastic waste
- Biomedical waste complaint
- Bulk waste

## Keywords

garbage, waste, dustbin, dumping, dirty street, cleaning, collection, rubbish, smell

Tamil/Tanglish:

kuppai, garbage edukala, kuppai podraanga, cleaning illa

## SLA

| Priority | SLA      |
|----------|----------|
| Critical | 4 hours  |
| High     | 12 hours |
| Medium   | 24 hours |
| Low      | 48 hours |

---

## 5. DEPARTMENT D003 — ROADS & PUBLIC WORKS

Department ID: D003

**Department Code:** ROADS\_PWD

## Categories

### Road Damage

- Potholes
- Damaged road
- Cracks
- Road collapse
- Unfinished road work

### Public Infrastructure

- Broken footpath
- Damaged bridge
- Damaged public building
- Damaged railing
- Missing road signs

### Construction

- Road construction
- Drain construction
- Public infrastructure maintenance

## Keywords

road, pothole, bridge, footpath, pavement, damaged, construction, accident

## SLA

| Priority | SLA      |
|----------|----------|
| Critical | 4 hours  |
| High     | 12 hours |
| Medium   | 48 hours |
| Low      | 7 days   |

## 6. DEPARTMENT D004 — ELECTRICITY & STREET LIGHTING

**Department ID:** D004

**Department Code:** ELECTRICITY

### Categories

#### Electricity

- Power outage
- Voltage fluctuation
- Transformer issue
- Electrical hazard
- Exposed electrical wire

#### Street Lighting

- Streetlight not working
- Multiple streetlights not working
- Damaged light pole
- Dark area

### Keywords

current, electricity, power, EB, wire, transformer, street light, light not working

### Priority Rules

#### Critical:

- Exposed live wire
- Electrical fire risk
- Transformer failure affecting critical infrastructure

#### High:

- Large-area outage
  - Multiple streetlights unavailable
-

## 7. DEPARTMENT D005 — PUBLIC TRANSPORT

**Department ID:** D005

**Department Code:** TRANSPORT

### Categories

- Bus delay
- Bus cancellation
- Route issue
- Bus overcrowding
- Driver/conductor complaint
- Bus stop issue
- Lost property
- Accessibility issue
- Fare-related complaint

### Metadata

Important fields:

- Route number
  - Vehicle number
  - Bus stop
  - Direction
  - Approximate time
  - Date
  - Passenger details if voluntarily provided
- 

## 8. DEPARTMENT D006 — TRAFFIC MANAGEMENT

**Department ID:** D006

**Department Code:** TRAFFIC

### Categories

- Traffic congestion
- Signal malfunction
- Illegal parking
- Road blockage
- Accident

- Traffic violation
- Missing traffic sign
- Unsafe junction

## High/Critical Examples

- Major accident
  - Signal failure at dangerous junction
  - Road completely blocked
  - Immediate public safety risk
- 

# 9. DEPARTMENT D007 — PUBLIC HEALTH

**Department ID:** D007

**Department Code:** HEALTH

## Categories

- Public health hazard
- Contaminated food complaint
- Mosquito breeding
- Unhygienic public area
- Public hospital/service issue
- Disease outbreak concern
- Medical facility complaint

## Important Design Rule

The platform should **not automatically make medical diagnoses**.

AI should classify and route the complaint.

Example:

Citizen reports suspected food poisoning.

AI:

**Category:** Public Health

**Priority:** High

**Department:** Public Health

**Action:** Route to appropriate health authority.

---



# 10. DEPARTMENT D008 — EMERGENCY & DISASTER MANAGEMENT

**Department ID:** D008

**Department Code:** DISASTER

## Categories

- Flood
- Cyclone
- Fire
- Landslide
- Building collapse
- Major accident
- Natural disaster
- Disaster warning
- Rescue request

## Priority

Almost all genuine emergency situations should receive:

**CRITICAL**

## Important

The system should recognize emergency keywords such as:

- trapped
- fire
- flood
- collapse
- injured
- life-threatening
- rescue
- unconscious
- major accident

The platform should display:

 **EMERGENCY — IMMEDIATE HUMAN REVIEW REQUIRED**

AI should **assist emergency routing, not replace emergency responders.**

---

## 11. DEPARTMENT D009 — POLICE & PUBLIC SAFETY

**Department ID:** D009

**Department Code:** POLICE

### Categories

- Crime report
- Theft
- Missing person
- Public disturbance
- Threat
- Domestic safety concern
- Suspicious activity
- Traffic safety
- Public harassment

### Critical

- Immediate threat to life
- Active violent incident
- Major accident
- Person in immediate danger

### System Behavior

The AI should identify emergency indicators and immediately escalate to human operators.

---

## 12. DEPARTMENT D010 — MUNICIPAL SERVICES

**Department ID:** D010

**Department Code:** MUNICIPAL

### Categories

- Public park issue
- Public toilet issue
- Encroachment
- Public property damage
- Civic facility complaint

- Street naming/signage
  - Public space maintenance
- 

## 13. DEPARTMENT D011 — ENVIRONMENT

**Department ID:** D011

**Department Code:** ENVIRONMENT

### Categories

- Air pollution
- Noise pollution
- Water pollution
- Illegal dumping
- Tree-related complaint
- Industrial pollution
- Environmental hazard

### Example

Citizen:

"There is a factory releasing black smoke every night."

AI:

```
Category: Environmental Pollution
Subcategory: Air Pollution
Priority: High
Department: Environment
Location: Extracted from call
```

---

## 14. DEPARTMENT D012 — REVENUE & CIVIC ADMINISTRATION

**Department ID:** D012

**Department Code:** REVENUE\_ADMIN

## Categories

### Property

- Property tax
- Property assessment
- Property records

### Certificates

- Birth certificate
- Death certificate
- Other civic certificates

### Municipal Administration

- Application status
- Document issue
- Service request
- General municipal inquiry

---

## 15. UNIVERSAL COMPLAINT METADATA

Every complaint in the database should contain standardized metadata.

### Identification Metadata

```
complaintId  
caseNumber  
callId  
citizenId  
departmentId  
categoryId
```

Example:

```
Complaint ID: CMP-2026-001248  
Call ID: CALL-2026-005821  
Citizen ID: CIT-004281  
Department: D001
```

## 16. CALL METADATA

Every incoming call should have:

```
callId  
callerId  
callStartTime  
callEndTime  
callDuration  
audioUrl  
language  
transcriptionStatus  
callStatus
```

Example:

```
Call ID: CALL-2026-005821  
Duration: 02:14  
Language: Tamil-English  
Status: Processed
```

---

## 17. CITIZEN METADATA

The platform should collect only information necessary for the service.

```
citizenId  
name  
phoneNumber  
preferredLanguage  
location  
city  
area  
pincode  
consent
```

For the hackathon, you can anonymize the phone number:

```
+91 *****4821
```

Do not display unnecessary personal information on dashboards.

---

## 18. LOCATION METADATA

This is extremely important for your project because you need **heat maps and hotspot detection**.

Store:

```
address
area
city
district
state
pincode
latitude
longitude
landmark
ward
zone
```

Example:

```
Area: Peelamedu
City: Coimbatore
District: Coimbatore
State: Tamil Nadu
Pincode: 641004
Latitude: XX.XXXX
Longitude: XX.XXXX
Ward: W-23
Zone: Central
```

For the prototype, you can use predefined coordinates for sample locations.

---

## 19. TRANSCRIPTION METADATA

The speech-to-text engine should produce:

```
transcript
detectedLanguage
```

```
languageConfidence
transcriptionConfidence
speakerCount
audioDuration
```

Example:

```
Detected Language:
Tamil-English

Language Confidence:
0.94

Transcript Confidence:
0.91
```

---

## 20. AI ANALYSIS METADATA

This is one of the most important parts of the database.

Store:

```
aiProcessed
modelName
modelVersion
processingTime
classification
classificationConfidence
summary
sentiment
sentimentConfidence
priority
priorityConfidence
departmentRecommendation
departmentConfidence
```

Example:

```
Classification:
Water Supply
```

Confidence:  
94%

Sentiment:  
Frustrated

Sentiment Confidence:  
89%

Department:  
Water Supply

Department Confidence:  
96%

---

## 21. COMPLAINT CLASSIFICATION METADATA

Use a hierarchy:

Department  
↓  
Category  
↓  
Subcategory  
↓  
Issue Type

Example:

Department:  
Water Supply & Sewerage

Category:  
Water Supply

Subcategory:  
No Water Supply

Issue Type:  
Area-wide disruption

This makes your analytics much more powerful.



---

## 22. PRIORITY METADATA

Store:

```
priority
priorityScore
priorityReason
emergencyFlag
riskLevel
```

Example:

```
Priority: HIGH
Score: 76
Emergency: false
Reason:
No water supply for three days affecting multiple households.
```

---

## 23. SENTIMENT METADATA

Possible values:

```
Positive
Neutral
Concerned
Frustrated
Angry
Distressed
Emergency
```

Example:

```
Sentiment:
Frustrated

Sentiment Score:
-0.72
```

Do not use sentiment alone to determine priority.

A calm citizen reporting a fire is still a critical complaint.

---

## 24. DUPLICATE DETECTION METADATA

Every duplicate check should store:

```
duplicateChecked  
duplicateComplaintId  
similarityScore  
matchingCategory  
matchingLocation  
duplicateStatus
```

Example:

```
Duplicate Checked:  
Yes  
  
Existing Complaint:  
CMP-2026-001122  
  
Similarity:  
91%  
  
Category Match:  
Yes  
  
Location Match:  
Yes  
  
Status:  
Possible Duplicate
```

---

## 25. SLA METADATA

Store:

```
slaDuration
slaStartTime
slaDueTime
remainingTime
slaStatus
escalationLevel
escalationTime
```

Example:

```
SLA:
12 hours

Created:
10:20 AM

Due:
10:20 PM

Status:
Approaching SLA

Escalation:
Level 1
```

---

## 26. COMPLAINT LIFECYCLE

Use a fixed status workflow:

```
RECEIVED
↓
AI_ANALYZED
↓
CLASSIFIED
↓
ASSIGNED
↓
IN_PROGRESS
↓
PENDING_INFORMATION
↓
RESOLVED
```

↓  
CLOSED

Possible alternative:

REOPENED

if the citizen reports that the issue was not actually resolved.

---

## 27. OFFICER METADATA

Each officer should have:

```
officerId
name
departmentId
designation
zone
contact
activeStatus
currentWorkload
averageResolutionTime
slaCompliance
complaintsResolved
```

Example:

```
Officer ID: OFF-1024
Department: Water
Zone: Central
Current Workload: 17
SLA Compliance: 94%
Average Resolution: 8.2 hours
```

---

## 28. DEPARTMENT PERFORMANCE METADATA

For analytics:

```
totalComplaints
openComplaints
resolvedComplaints
criticalComplaints
slaBreaches
averageResolutionTime
slaCompliancePercentage
citizenSatisfaction
```

Example:

Water Department

Total: 1,248

Open: 342

Resolved: 906

SLA Compliance: 87%

Average Resolution: 9.4 hours

---

## 29. NOTIFICATION METADATA

Store:

```
notificationId
recipientId
complaintId
notificationType
message
timestamp
readStatus
```

Notification types:

- Complaint created
- Complaint assigned
- SLA approaching
- SLA breached
- Complaint escalated
- Complaint resolved
- Duplicate detected
- Emergency detected

---

## 30. AI RECOMMENDATION METADATA

This is useful for your **Officer Copilot**.

Store:

```
recommendationId  
complaintId  
recommendationType  
recommendationText  
confidence  
generatedAt  
acceptedByOfficer  
officerAction
```

Example:

```
Recommendation:  
"Dispatch a field inspection team."  
  
Confidence:  
91%  
  
Officer:  
Accepted  
  
Action:  
Inspection Assigned
```

---

## 31. PREDICTIVE ANALYTICS METADATA

For recurring issue prediction:

```
predictionId  
location  
department  
category  
timeWindow  
historicalComplaintCount  
predictedComplaintCount
```

```
growthRate
riskLevel
recommendedAction
```

Example:

```
Area:
Peelamedu

Category:
Water

Historical:
184 complaints

Predicted:
211 complaints

Expected Increase:
14.7%

Risk:
High

Recommendation:
Inspect water distribution network.
```

---

## 32. HEAT MAP METADATA

For geographic analytics:

```
locationId
latitude
longitude
area
complaintCount
criticalCount
dominantCategory
trend
riskScore
```

Example:

Area: Peelamedu  
Complaints: 124  
Critical: 18  
Dominant Category: Water  
Trend: Increasing  
Risk Score: 82

## 33. COMPLETE COMPLAINT OBJECT

A simplified complaint record could look like this:

```
{
  "complaintId": "CMP-2026-001248",
  "callId": "CALL-2026-005821",

  "citizen": {
    "citizenId": "CIT-004281",
    "language": "ta-en",
    "location": {
      "area": "Peelamedu",
      "city": "Coimbatore",
      "state": "Tamil Nadu"
    }
  },

  "call": {
    "duration": 134,
    "audioUrl": "call_5821.wav",
    "language": "Tamil-English"
  },

  "transcription": {
    "text": "Enga area-la moonu naala water varala...",
    "confidence": 0.91
  },

  "classification": {
    "department": "D001",
    "category": "Water Supply",
    "subcategory": "No Water Supply",
    "confidence": 0.94
  },
}
```



```

"aiAnalysis": {
  "summary": "Water supply disruption for three days.",
  "sentiment": "Frustrated",
  "priority": "HIGH",
  "priorityScore": 76
},

"duplicateDetection": {
  "checked": true,
  "possibleDuplicate": true,
  "complaintId": "CMP-2026-001122",
  "similarity": 0.91
},

"sla": {
  "durationHours": 12,
  "status": "APPROACHING",
  "escalationLevel": 1
},

"resolution": {
  "status": "ASSIGNED",
  "assignedOfficer": "OFF-1024"
}
}

```

## 34. DEPARTMENT MASTER OBJECT

Each department can have a standard record:

```

{
  "departmentId": "D001",
  "departmentCode": "WATER",
  "name": "Water Supply & Sewerage",
  "domain": "Public Utilities",

  "categories": [
    {
      "categoryId": "W001",
      "name": "Water Supply",
      "subcategories": [
        "No Water Supply",
        "Low Pressure",
        "Contaminated Water",

```

```

        "Irregular Supply"
    ]
},
{
    "categoryId": "W002",
    "name": "Water Leakage",
    "subcategories": [
        "Pipeline Leakage",
        "Pipeline Burst",
        "Connection Leakage"
    ]
}
],

"sla": {
    "critical": 4,
    "high": 12,
    "medium": 24,
    "low": 72
},

"keywords": [
    "water",
    "pipeline",
    "leakage",
    "tanker",
    "drainage",
    "thanni",
    "water varala"
],

"active": true
}

```

## 35. MASTER METADATA CATEGORIES

Your entire system can therefore be organized around these metadata groups:

1. Citizen Metadata
2. Call Metadata
3. Audio Metadata
4. Transcription Metadata
5. Language Metadata
6. Complaint Metadata

7. Department Metadata
8. Category Metadata
9. Location Metadata
10. AI Classification Metadata
11. Sentiment Metadata
12. Priority Metadata
13. Duplicate Detection Metadata
14. SLA Metadata
15. Officer Metadata
16. Assignment Metadata
17. Resolution Metadata
18. Notification Metadata
19. Prediction Metadata
20. Analytics Metadata
21. Audit Metadata

## 36. AUDIT METADATA

For an enterprise/government system, every important action should be traceable.

Store:

```
auditId
userId
role
action
complaintId
timestamp
previousValue
newValue
```

Example:

Officer OFF-1024

Action:  
Changed status

Previous:  
ASSIGNED

New:

IN\_PROGRESS

Time:

14 Aug 2026, 14:32

This gives you **accountability**, which directly supports one of the expected outcomes in the problem statement.

---

## 37. ROLE METADATA

Your system can have four roles.

### CITIZEN

Can:

- submit complaint
- track complaint
- receive updates
- provide feedback

### CALL CENTER AGENT

Can:

- receive calls
- view transcription
- verify AI classification
- create complaint
- modify incorrect metadata

### DEPARTMENT OFFICER

Can:

- view assigned complaints
- update status
- add resolution
- escalate
- view department analytics

## ADMINISTRATOR

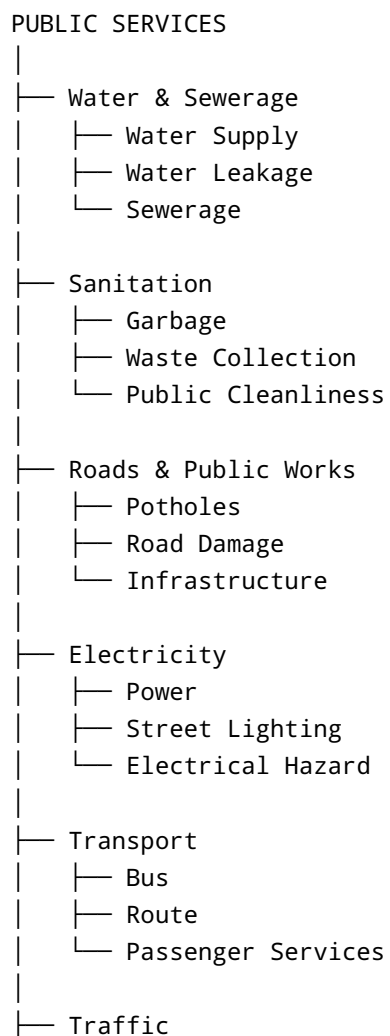
Can:

- manage departments
- manage officers
- view all complaints
- view city-wide analytics
- configure SLA
- monitor performance

---

## 38. FINAL DEPARTMENT TAXONOMY

For your prototype, the hierarchy should look like:



- | | | Congestion
- | | | Signals
- | | | Road Safety
- | |
- | | Public Health
- | | | Health Hazard
- | | | Hygiene
- | | | Public Facilities
- | |
- | | Emergency
- | | | Fire
- | | | Flood
- | | | Accident
- | | | Disaster
- | |
- | | Police & Safety
- | | | Crime
- | | | Missing Person
- | | | Public Safety
- | |
- | | Municipal Services
- | | | Parks
- | | | Public Facilities
- | | | Encroachment
- | |
- | | Environment
- | | | Air Pollution
- | | | Noise Pollution
- | | | Water Pollution
- | | | Trees
- | |
- | | Revenue & Administration
- | | | Property Tax
- | | | Certificates
- | | | Civic Applications

---

## 39. THE MOST IMPORTANT METADATA FOR YOUR DEMO

If you're running out of development time, **these are the fields you absolutely need:**

## **Call**

- Call ID
- Audio
- Transcript
- Language
- Duration

## **Complaint**

- Complaint ID
- Summary
- Category
- Subcategory
- Location
- Department
- Priority
- Sentiment

## **AI**

- Confidence
- Department recommendation
- Duplicate similarity
- AI recommendation

## **SLA**

- SLA duration
- Due time
- Current status
- Escalation level

## **Officer**

- Assigned officer
- Department
- Zone

## **Analytics**

- Complaint count
- Resolution rate
- SLA compliance
- Critical complaints
- Location
- Category trend

These fields alone are sufficient to make your prototype feel like a **complete enterprise platform**.

---

## 40. THE CORE IDEA TO REMEMBER

Your platform isn't simply storing complaints.

It is creating a **Citizen Call Intelligence Record**.

The transformation should be:

UNSTRUCTURED

"I called because there has been no water  
in our area for three days..."

↓

STRUCTURED INTELLIGENCE

|                    |                           |
|--------------------|---------------------------|
| Department         | → Water Supply            |
| Category           | → No Water Supply         |
| Location           | → Peelamedu               |
| Language           | → Tamil-English           |
| Sentiment          | → Frustrated              |
| Priority           | → High                    |
| Duplicate          | → 91% match               |
| SLA                | → 12 hours                |
| Officer            | → OFF-1024                |
| Recommended Action | → Field inspection        |
| Prediction         | → Rising water complaints |

That is the fundamental value proposition of your project:

**"We transform citizen conversations into structured, actionable governance intelligence."**

And this metadata architecture gives you the foundation for **AI classification, department routing, duplicate detection, SLA tracking, officer accountability, heat maps, predictive analytics and executive dashboards** without needing to build every feature as a separate system.